

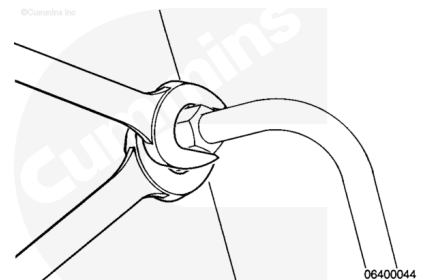
Trainings ▾

Remove

Rail Applications

⚠ WARNING ⚠

Fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.

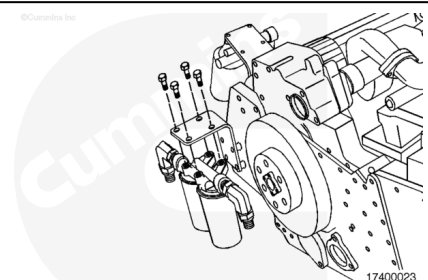


Note : Use two wrenches to remove the fuel hoses.

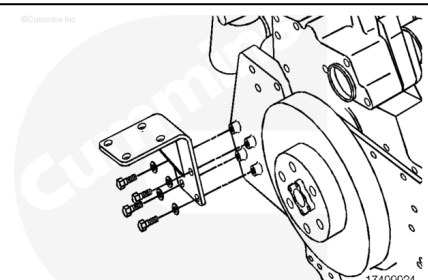
Support the mating fittings with a wrench. Loosen the fuel hose nuts with the other wrench.

Remove the inlet hose, outlet hose, and o-rings from the fuel filter head.

Remove the four capscrews and the fuel filter head from the bracket.



Remove the four mounting capscrews and the bracket from the lubricating oil pan.

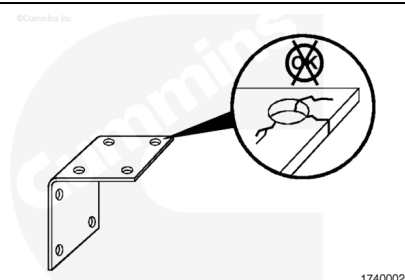


Inspect for Reuse

Rail Applications

Fuel Filter Head Bracket - Inspect for Reuse

Check for cracks in the bracket.

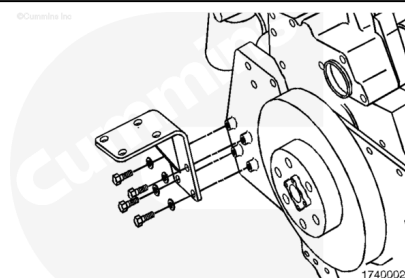


Install

Rail Applications

⚠ WARNING ⚠

Fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.

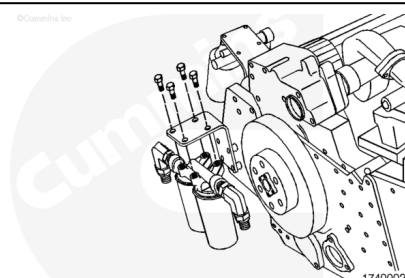


Install the bracket and four capscrews on the lubricating oil pan. Tighten the capscrews.

Torque Value: 75 n•m [55 ft-lb]

Install the fuel filter head and four capscrews on the bracket. Tighten the capscrews.

Torque Value: 25 n•m [221 in-lb]

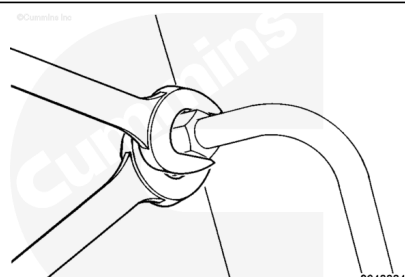


Install the two hoses and o-rings on the fuel filter head.

Tighten the fittings.

Torque Value:

Fitting - Fuel tank to fuel filter 115 n•m [85 ft-lb]



Torque Value:

Fitting - Fuel filter to fuel pump 90 n•m [66 ft-lb]

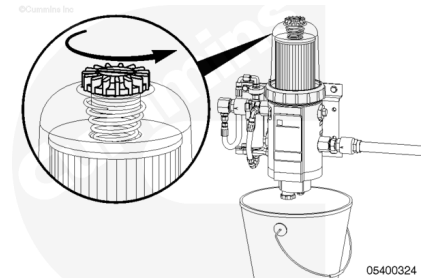
Operate the engine and check for leaks.

Drain

with Electronically Actuated Injector

⚠ WARNING ⚠

Depending on the circumstance, fuel is flammable. When inspecting or performing service or repairs on the fuel system, to reduce the possibility of fire and resulting severe personal injury, death or property damage, never smoke or allow sparks or flames (such as pilot lights, electrical switches, or welding equipment) in the work area.



⚠ CAUTION ⚠

Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. The fuel must be disposed of in accordance with local environmental regulations.

Industrial and Power Generation Application

- Accumulated water **must** be drained daily.
- For fuel that has a high water content, more frequent draining will be necessary.

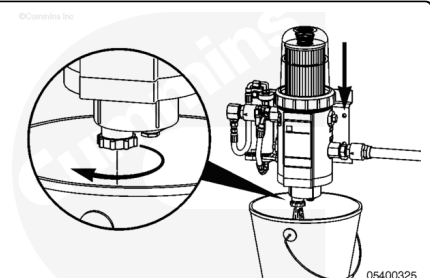
Shut the engine OFF.

Place a suitable container under the fuel filter.

With the fuel supply valve closed, open the vent cap to break the air lock in the filter.

⚠ CAUTION ⚠

Do not spill or drain fuel when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. The fuel must be disposed of in accordance with local environmental regulations.



Open the drain valve. If there is accumulated water in the system, the water will flow out first.

When fuel begins to flow out of the drain, close the drain valve.

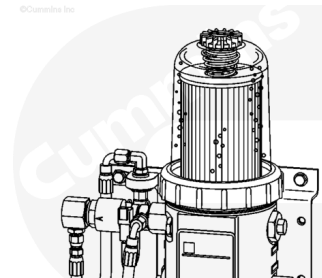
The drained liquids **must** be disposed of in accordance with local environmental regulations.

Close the vent cap.

Start the engine, and raise the rpm for 1 minute to purge the air from the system.

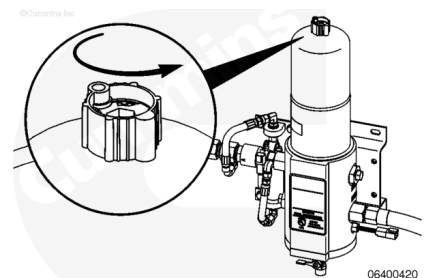
If the engine fails to start, prime the filter to remove the excess air. Refer to Procedure 006-015 in Section 6.

(/qs3/pubsys2/xml/en/procedures/20/20-006-015-tr.html)



⚠ WARNING ⚠

Depending on the circumstance, fuel is flammable. When inspecting or performing service or repairs on the fuel system, to reduce the possibility of fire and resulting severe personal injury, death or property damage, never smoke or allow sparks or flames (such as pilot lights, electrical switches, or welding equipment) in the work area.



⚠ CAUTION ⚠

Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. The fuel must be disposed of in accordance with local environmental regulations.

Marine application engines are equipped with two different Stage 1 filter systems. They are identified as Duplex (filters with individual shutoff valves) and Single (filters without individual shutoff valves).

Duplex filter applications enable the engine to remain in operation while the filters are changed.

Single filter applications require the engine to be shut down for service.

Marine Single Filter Application

- The accumulated water **must** be drained daily.
- For fuel that has a high water content, more frequent draining will be necessary.

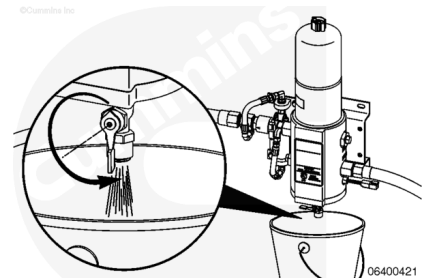
Shut the engine OFF.

Place a suitable container under the fuel filter.

With the fuel supply valve closed, open the vent cap to break the air lock in the filter.

⚠ CAUTION ⚠

Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. The fuel must be disposed of in accordance with local environmental regulations.



Open the drain valve. If there is accumulated water in the system, the water will flow out first.

When fuel begins to flow out of the drain, close the drain valve.

The drained liquids **must** be disposed of in accordance with local environmental regulations.

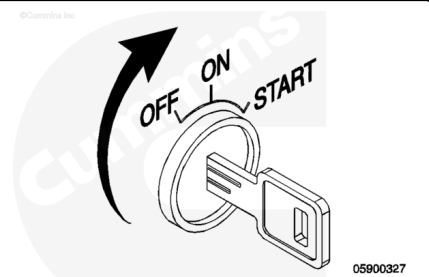
Close the vent cap.

Start the engine, and raise the rpm for 1 minute to purge the air from the system.

If the engine does **not** start, perform the prime procedure to remove the excess air.

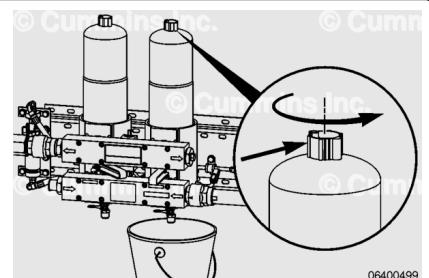
If the engine fails to start, prime the filter to remove the excess air. Refer to Procedure 006-015 in Section 6.

(/qs3/pubsys2/xml/en/procedures/20/20-006-015-tr.html)



⚠ WARNING ⚠

Depending on the circumstance, fuel is flammable. When inspecting or performing service or repairs on the fuel system, to reduce the possibility of fire and resulting severe personal injury, death or property damage, never smoke or allow sparks or flames (such as pilot lights, electrical switches, or welding equipment) in the work area.



⚠ CAUTION ⚠

Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. The fuel must be disposed of in

accordance with local environmental regulations.

Marine Duplex Filter Application

- The accumulated water **must** be drained daily.
- For fuel that has a high water content, more frequent draining will be necessary.

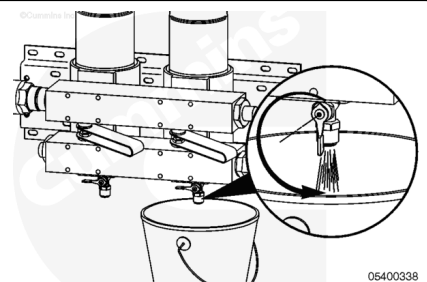
Move the filter selector valve handle to the OFF position for the filter to be drained.

Place a suitable container under the fuel filter.

With the filter selector valve closed, open the vent cap to break the air lock in the filter.

⚠ CAUTION ⚠

Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. The fuel must be disposed of in accordance with local environmental regulations.



Open the drain valve. If there is accumulated water in the system, the water will flow out first.

When fuel begins to flow out of the drain, close the drain valve.

The drained liquids **must** be disposed of in accordance with local environmental regulations.

Close the vent cap.

If the engine was shut OFF:

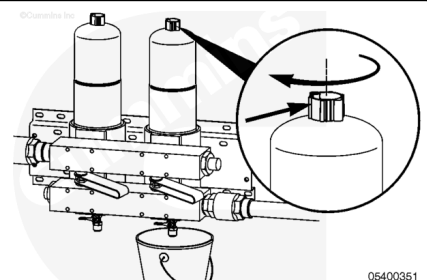
Start the engine, and raise the rpm for 1 minute to purge the air from the system.

If the engine is in operation, slowly open the selector valve handle to the ON position, allowing extra time to purge air from the system.

If the engine fails to start, perform the prime procedure to remove the excess air.

If the engine fails to start, prime the filter to remove the excess air. Refer to Procedure 006-015 in Section 6.

(/qs3/pubsys2/xml/en/procedures/20/20-006-015-tr.html)

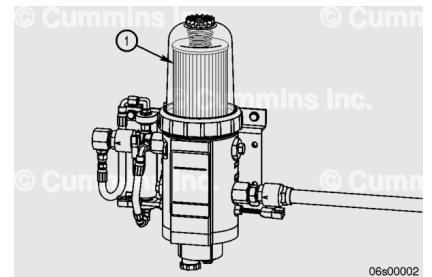


Maintenance Check

with Electronically Actuated Injector

Industrial and Power Generation Applications

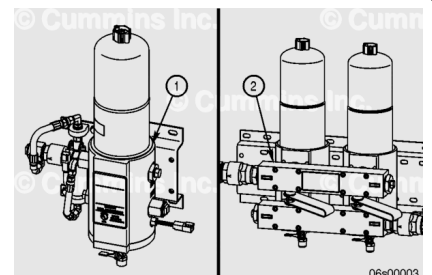
- Industrial and Power Generation application stage 1 filters contain a clear cover for inspection of the fuel level in the filter. Over time the fuel level in the filter will rise as the filter becomes contaminated.
- The filter **must** be replaced if the fuel level reaches the top of the filter element (1) prior to the standard maintenance interval. Check the fuel level in the filter. Replace if required. Refer to Procedure 006-075 in Section 4. (</qs3/pubsys2/xml/en/procedures/20/20-006-075-om.html>)



Note : The fuel level in the filter is **not** to be used as an extended life monitor. Filter life **must not** exceed the standard maintenance interval.

Marine Applications

- Marine application engines are equipped with two different filter systems. They are identified as Non-Classed (1) (filters without individual shutoff valves) and Classed (2) (filters with individual shutoff valves).



The filter element in these filter systems use a steel cover and therefore contains a pop-up indicator to indicate when the filters are contaminated before the standard maintenance interval. The pop up indicator is located on the right side of the filter housing on Non-Classed systems and between the two filters on Classed systems.

Note : The pop-up indicators are **not** shown on these graphics.

When the indicator pops up, all filters **must** be changed. Refer to Procedure 006-075 in Section 6.

(</qs3/pubsys2/xml/en/procedures/20/20-006-075-tr.html>)

Note : The pop up indicator is **not** to be used as an extended life monitor. Filter life **must not** exceed the standard maintenance interval.

